F_2-slot

① $\boxed{10m}$

$X \equiv 1 \pmod 3$

$X \equiv 2 \pmod 4$

$X \equiv 3 \pmod 5$

$a=1,\ b=2,\ c=3,\ M = 3\cdot4\cdot5 = 60$

$M_1 = \frac{60}{3} = 20 \qquad M_1^{-1} = M_1 \cdot m_1^{-1} \bmod 60$

$M_2 = \frac{60}{4} = 15 \qquad M_1^{-1} = 2$

$M_3 = \frac{60}{5} = 12 \qquad M_2^{-1} = 3$

$M_3^{-1} = 3$

$X = (1\cdot20\cdot2 + 2\cdot15\cdot3 + 3\cdot12\cdot3) \bmod 60$

$a \cdot M_1 \cdot m_1^{-1}$

$= (40 + 90 + 108) \bmod 60$

$X = 238 \bmod 60 = 58$

$\boxed{X = 58}$ $\boxed{4M}$

② ⓐ $a^{p-1} \equiv 1 \pmod p$ $\boxed{4M}$

If a is an Integer and p is a prime number, then $(a^p - a)$ is a multiple of ~~a~~ p

20^{-1} $20 \cdot x \bmod 26 = 1$

Consider the set of residues

① $S = \{1, 2, 3, \ldots p-1\}$, all the elements are relatively prime to p.

② multiply a

$S' = \{a\cdot1, a\cdot2, a\cdot3, \ldots a\cdot(p-1)\}$

Each element in S' is congruent modulo p to some element in S, and the elements remain distinct

$1\cdot2\cdot3\ldots p-1 \equiv a^{p-1}(1\cdot2\cdot3 \ldots p-1) \pmod p$

$(p-1)!$ cancel out $\Rightarrow$ $a^{p-1} \equiv 1 \pmod p$

$3^{12} \bmod 11 \Rightarrow 3^{10}\cdot3^2 \bmod 11 \Rightarrow (3^{10} \bmod 11)\cdot(3^2 \bmod 11) \bmod 11$

$\Rightarrow 1 \cdot 9 \bmod 11 = 9$ $\boxed{2M}$

② ⓑ Definition - Euler's totient fun

$\phi(n) = p \cdot q$ for $n = p \cdot q$ where

$\phi(n) = |\{x \mid 1 \le x \le n, \gcd(x, n) = 1\}|$

proof:

$\phi(n) = (p-1)(q-1),\ n = pq$

③ Hill cipher ciphertext = QVNAYQHI

16 | 21 | 13 | 0 | 24 | 16 | 7 | 8

Key matrix $K = \begin{bmatrix} 2 & 3 \\ 7 & 8 \end{bmatrix} \Rightarrow K^{-1} = \frac{1}{\det K} \cdot \text{adj}\,K$

$K^{-1} = \begin{bmatrix} 2 & 3 \\ 7 & 8 \end{bmatrix}^{-1} = \det K = 16 - 21 = -5$

$26 - (5 \bmod 26) = 26 - 5 = 21$

$\text{adj}\,K = \begin{bmatrix} 8 & -3 \\ -7 & 2 \end{bmatrix} \Rightarrow K^{-1} = \frac{1}{-5}\begin{bmatrix} 8 & -3 \\ -7 & 2 \end{bmatrix}$

$K^{-1} = \begin{bmatrix} -\frac{8}{5} & \frac{3}{5} \\ \frac{7}{5} & -\frac{2}{5} \end{bmatrix} \bmod 26$

$= \begin{bmatrix} -8\cdot5^{-1} \bmod 26 & 3\cdot5^{-1} \bmod 26 \\ 7\cdot5^{-1} \bmod 26 & -2\cdot5^{-1} \bmod 26 \end{bmatrix}$

$= \begin{bmatrix} -168 \bmod 26 & 63 \bmod 26 \\ 147 \bmod 26 & -42 \bmod 26 \end{bmatrix}$

$\begin{bmatrix} 26-12 & 11 \\ 17 & 26-(42 \bmod 26) \end{bmatrix} = \begin{bmatrix} 14 & 11 \\ 17 & 10 \end{bmatrix}$

$5^{-1} x \bmod 26 = 1 = 21$

$\frac{21\cdot8}{168},\ \frac{21\cdot3}{63},\ \frac{21\cdot7}{147}$

$\boxed{K^{-1} = \begin{bmatrix} 14 & 11 \\ 17 & 10 \end{bmatrix}}$

$\begin{bmatrix} 14 & 11 \\ 17 & 10 \end{bmatrix}_{2\times2} \cdot \begin{bmatrix} Q \\ V \end{bmatrix} = \begin{bmatrix} 16 \\ 21 \end{bmatrix}_{2\times1} = \begin{bmatrix} 14\times16 + 11\cdot21 \\ 17\cdot16 + 10\cdot21 \end{bmatrix} = \begin{bmatrix} 224+231 \\ 272+210 \end{bmatrix}$

$\begin{bmatrix} N \\ A \end{bmatrix} = \begin{bmatrix} 13 \\ 0 \end{bmatrix} = \begin{bmatrix} 14\cdot13 + 11\cdot0 \\ 17\cdot13 + 10\cdot0 \end{bmatrix} = \begin{bmatrix} 182 \\ 221 \end{bmatrix}$

$\begin{bmatrix} Y \\ Q \end{bmatrix} = \begin{bmatrix} 24 \\ 16 \end{bmatrix} = \begin{bmatrix} 14\cdot24 + 11\cdot16 \\ 17\cdot24 + 10\cdot16 \end{bmatrix} = \begin{bmatrix} 336 + 176 \\ 408 + 160 \end{bmatrix}$

$\begin{bmatrix} H \\ I \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 14\cdot7 + 11\cdot8 \\ 17\cdot7 + 10\cdot8 \end{bmatrix} = \begin{bmatrix} 98+88 \\ 119+80 \end{bmatrix}$

$$\begin{bmatrix} 455 \\ 482 \end{bmatrix} \bmod 26 = \begin{bmatrix} 13 \\ 14 \end{bmatrix} \cdot \begin{bmatrix} N \\ O \end{bmatrix}$$

$$\begin{bmatrix} 182 \\ 221 \end{bmatrix} \bmod 26 = \begin{bmatrix} 0 \\ 13 \end{bmatrix} \begin{bmatrix} A \\ N \end{bmatrix}$$

$$\begin{bmatrix} \overset{512}{\cancel{402}} \\ 568 \end{bmatrix} \bmod 26 = \begin{bmatrix} \overset{18}{\cancel{12}} \\ 22 \end{bmatrix} \begin{bmatrix} M \\ W \end{bmatrix} S$$

$$\begin{bmatrix} 186 \\ 199 \end{bmatrix} \bmod 26 = \begin{bmatrix} 4 \\ 17 \end{bmatrix} \begin{bmatrix} E \\ R \end{bmatrix}$$

NOANMWER

$\boxed{4M}$

(2) (b) There are total numbers 1 to pq $(n=pq)$

we consider as per the definitions,

the numbers which are divisible p & q are

$(p-1) \cdot q$ terms and $(q-1) \cdot p$ terms

Total no. of elements divisible by either p or q

$$q+p-1$$

$$\phi(pq) = pq - (q+p-1)$$
$$= pq - q - p + 1$$
$$= p(q-1) - q + 1$$
$$= p(q-1) - (q-1)$$
$$\phi(pq) = (q-1)(p-1)$$

(4)

$L_2 = 0011 = (3)_{10}$ $\quad R_2 = 1111 = (15)_{10}$

Round 2

$f_2 (2,$

$L_n = R_{n-1}$

$R_n = L_{n-1} \oplus f_n(R_{n-1}, K_n)$

$L_n = R_{n-1} \oplus f_{n}(R_{n-1}, K)$

$R_n = L_{n-1}$

$f_n(x, K) = (2 \cdot n \cdot K)^x \bmod 15$

$\cancel{A\,2\,x}$

$f_i(x, K) = (2 \cdot i \cdot K)^x \bmod 15$

Round-2 :-

$f_2 (2 \cdot 2 \cdot 7)^3 \bmod 15 = 28^3 \bmod 15 = 7$

$L_1 \Rightarrow f = 0111 \oplus R_2 \Rightarrow 0111 \oplus 1111 = 1000$

$L_2 \quad R_2 \oplus f$

$L_1 = 1000 \quad R_1 = 0011$

Round-1

$f_1(L_1, K_1) = f(1000, \cancel{7}) = (2 \cdot 1 \cdot 7)^8 \bmod 15$
(8)

$= 14^8 \bmod 15$

$= \cancel{0001 \oplus}$

$= 1 \oplus R_1$

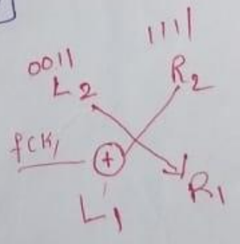
$= 0001 \oplus 0011$

$= 0010$

L_1

$L_0 = 0010 \quad R_0 = 1000$

(5) AES key gen, ENC & DEC $\boxed{10M}$



$R_1 = L_2$

$L_1 = R_2 \oplus f(R_1, K)$

(1)

$$x \equiv 1 \pmod 3$$
$$x \equiv 2 \pmod 4$$
$$x \equiv 3 \pmod 5$$

$M = 3 \times 4 \times 5 = 60$

$M_1 = 60/3 = 20$, $M_2 = 60/4 = 15$, $M_3 = 60/5 = 12$

$M_1^{-1} = 2$ $M_2^{-1} = 3$ $M_3^{-1} = 3$

$x = (1 * 20 * 2 + 3 * 15 * 3 + 3 * 12 * 3) \bmod 60$

$= (40 + 90 + 108) \bmod 60$

$= (238) \bmod 60$

$= \boxed{58}$

238
180
58

(2) Fermat's Statement & Prove

$3^{12} \bmod 11 = (3^{11} \times 3^{1}) \bmod 11$

$= [(3^{11} \bmod 11) \times (3^{1} \bmod 11)] \bmod 11$

$= (3 \times 3) \bmod 11$

$= \boxed{9}$

(b) Euler's theorem definition

There are total numbers 1 to pq

~~We count~~

We consider as per definition

The numbers which are divisible p & q are

$(p-1)\, q$ terms and $(q-1)\, p$ terms

total No. of elements divisible by either p & q

$(q+p-1)$

$$\phi(pq) = pq - (q+p-1)$$

$$= pq - q - p + 1$$

$$\phi(n) = (p-1)(q-1)$$

K^{-1}

$$K^{-1} = \begin{bmatrix} 2 & 3 \\ 7 & 8 \end{bmatrix}^{-1}$$

$=$

$\det(K) = -5$

$= 21$

$$= \frac{1}{21}\begin{bmatrix} 8 & -3 \\ -7 & 2 \end{bmatrix}$$

$$= 5\begin{bmatrix} 8 & -3 \\ -7 & -2 \end{bmatrix} = \begin{bmatrix} 40 & -15 \\ -35 & -10 \end{bmatrix} \bmod 26$$

-9
26

$$= \begin{matrix} 14 & 11 \\ 9 & 16 \end{matrix}$$

$-26/2$

$$= 5\begin{bmatrix} 8 & 23 \\ 19 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 40 & 115 \\ 95 & 10 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 14 & 11 \\ 17 & 10 \end{bmatrix}$$

2×2
2×4

$$P = K^{-1}C = \begin{bmatrix} 14 & 11 \\ 17 & 10 \end{bmatrix} \times \begin{bmatrix} 16 & 13 & 24 \\ 21 & 0 & 16 \end{bmatrix}$$

$$= \begin{bmatrix} 13 & 0 & 18 & 4 \\ 14 & 13 & 22 & 17 \end{bmatrix}$$

$=$ ~~N O~~ A N $\boxed{\text{NO ANSWER}}$

(4)

$L_2 = 0011 = 3$ $R_2 = 1111 = 15$

Round 2

$i = 2$

$f = (2, 2, 7)$

$f(0011, 7) = (28)^3 \bmod 15$

$= 7$

$= 0111 \oplus 1111$

$= 1000$

Swap

$L_1 = 1000$ $R_1 = 0011$

Round 1 ($i = 1$)

$f(1000, 7) = (14)^8 \bmod 15$

$= 0001 \oplus 0011$

$= 0010$

Swap

0010 1000 = Plain text

(5) AES - Key Generation & Encry / Decry